Agent

An agent can be anything that perceive its environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving**, **thinking**, and **acting**. An agent can be:

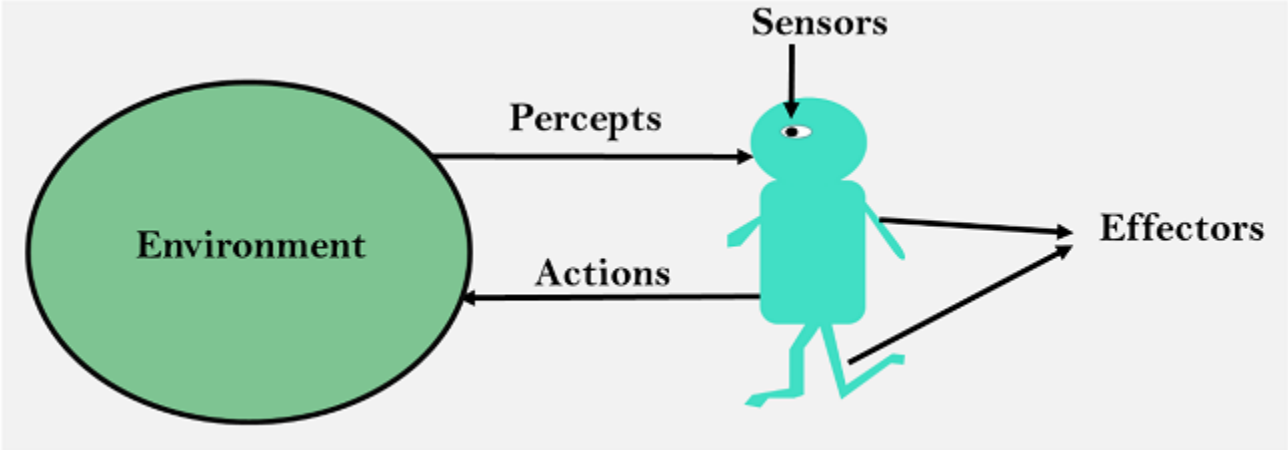
- **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

Hence the world around us is full of agents such as thermostat, cellphone, camera, and even we are also agents. Before moving forward, we should first know about sensors, effectors, and actuators.

Sensor: Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

Actuators: Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

Effectors: Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.



Intelligent Agents:

An intelligent agent is an autonomous entity which act upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Following are the main four rules for an AI agent:

- **Rule 1:** An AI agent must have the ability to perceive the environment.
- **Rule 2:** The observation must be used to make decisions.
- **Rule 3:** Decision should result in an action.
- **Rule 4:** The action taken by an AI agent must be a rational action.

Rational Agent:

A rational agent is an agent which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions.

A rational agent is said to perform the right things. AI is about creating rational agents to use for game theory and decision theory for various real-world scenarios.

For an AI agent, the rational action is most important because in AI reinforcement learning algorithm, for each best possible action, agent gets the positive reward and for each wrong action, an agent gets a negative reward.

Rationality:

The rationality of an agent is measured by its performance measure. Rationality can be judged on the basis of following points:

- Performance measure which defines the success criterion.
- Agent prior knowledge of its environment.
- Best possible actions that an agent can perform.
- The sequence of percepts.

Structure of an AI Agent

The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

Agent = Architecture + Agent program

Following are the main three terms involved in the structure of an AI agent:
Architecture: Architecture is machinery that an AI agent executes on.
Agent Function: Agent function is used to map a percept to an action.

$f : P^* \longrightarrow A$

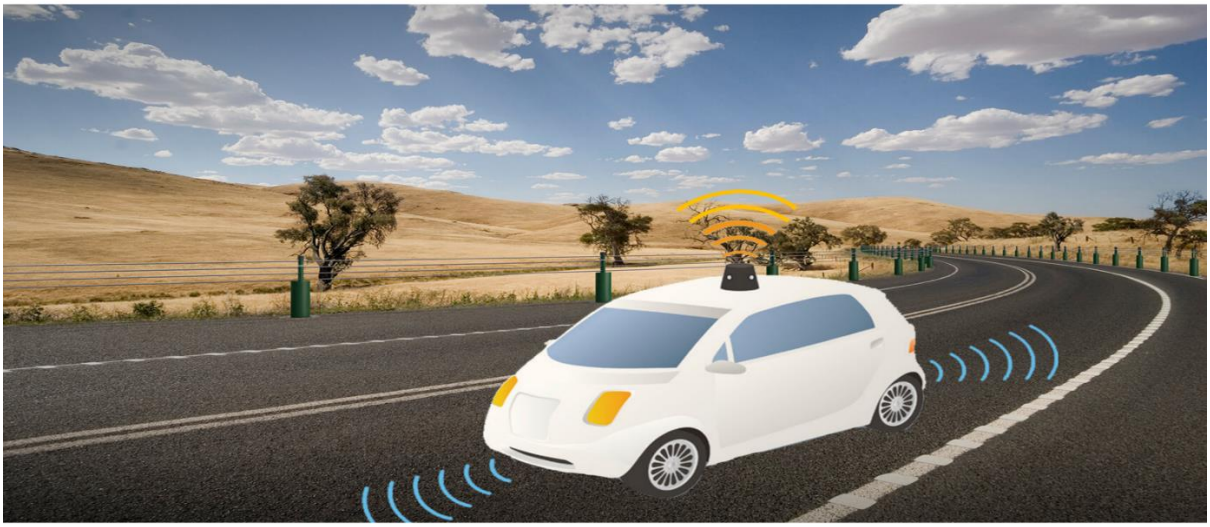
Agent program: Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function f.

PEAS Representation

PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:

- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

Here performance measure is the objective for the success of an agent's behavior.
PEAS for self-driving cars:



Let's suppose a self-driving car then PEAS representation will be:

Performance: Safety, time, legal drive, comfort

Environment: Roads, other vehicles, road signs, pedestrian

Actuators: Steering, accelerator, brake, signal, horn

Sensors: Camera, GPS, speedometer, odometer, accelerometer, sonar.

Example of Agents with their PEAS representation

Agent	Performance measure	Environment	Actuators	Sensors
1. Medical Diagnose	○ Healthy patient ○ Minimized cost	○ Patient ○ Hospital ○ Staff	○ Tests ○ Treatments	Keyboard (Entry of symptoms)
2. Vacuum Cleaner	○ Cleanness ○ Efficiency ○ Battery life ○ Security	○ Room ○ Table ○ Wood floor ○ Carpet ○ Various obstacles	○ Wheels ○ Brushes ○ Vacuum Extractor	○ Camera ○ Dirt detection sensor ○ Cliff sensor ○ Bump Sensor ○ Infrared Wall Sensor
3. Part - picking Robot	○ Percentage of parts in correct bins.	○ Conveyor belt with parts, ○ Bins	○ Jointed Arms ○ Hand	○ Camera ○ Joint angle sensors.

Agent Architecture:

Agent architecture in artificial intelligence (AI) is the underlying framework that defines how AI agents perceive their environment, make decisions, and take actions to achieve their goals. These architectures enable agents to act autonomously in a wide range of scenarios, from simple tasks to complex and dynamic environments. Understanding agent architecture is essential for developing sophisticated AI systems capable of adaptive learning, problem-solving, and decision-making.

AI Agent Architectures examine the complex structures that shape how machines perceive, reason, and act in their environments in the pursuit of autonomous intelligence. This article explores the various structures that shape AI's decision-making capabilities

AI Agent Architecture

An intelligent agent system's basic components and interactions are outlined in an AI agent architecture, which functions as a conceptual design. It offers a methodical framework for creating, putting into practice, and comprehending agents that may independently interact with their surroundings to accomplish predetermined goals. Agent architectures are important because they provide a methodical way to design and evaluate complex AI systems, enabling scientists and engineers to build agents with certain features and functions.

Let's discuss each component of AI Agent architecture in detail:

1. Profiling Module: The Eyes and Ears of the Agent

- The agent is endowed with sensory expertise via the profiling module, it is also known as the **perception module**. With the help of this module, the agent may collect and analyze information from its environment like how human senses work. It interprets unprocessed sensory data, eliminates extra/redundant details, and extracts important characteristics that are essential for making decisions. The agent's accurate and thorough grasp of its surroundings is ensured by the

profile module, which helps it comprehend visual signals, recognize speech patterns, and sense tactile inputs.

- Imagine a self-driving automobile that has radar, lidar, and camera sensors. Through the processing of data from various sensors, the profile module recognizes barriers, traffic lights, and lane markers, allowing the vehicle to go safely.

2. Memory Module: The Repository of Knowledge

- The memory module is essential for organizing and storing data as it serves as the **agent's knowledge base**. It allows the agent to remember information, rules, patterns, and events from the past. It functions similarly to the agent's short and long-term memory. For the agent to make wise judgements, learn from its interactions, and have a sophisticated grasp of its surroundings, this module is crucial.
- To provide timely and correct replies during discussions, a virtual assistant chatbot, for example, uses its memory module to remember client requests, product specification, and commonly asked questions.

3. Planning Module: The Strategist behind Decisions

- The agent's **central decision-making command post** for making decisions and acting in a goal-oriented manner is the planning module. This module examines the present situation and determines the best course of action to accomplish the agent's goals, taking into account information from the memory and profiling modules. Using strategies like optimization approaches, search algorithms, or rule-based systems, the planning module figures out the best course of action.
- For instance, a delivery routing agent utilizes the planning module to optimize the most effective routes and timetables for its fleet by taking into account variables like vehicle capacity, traffic patterns, and delivery priority.

4. Action Module: From Plans to Execution

- The choices and plans established by the planning module are implemented by the action module. It converts plans into executable commands and serves as the agent's interface with the outside world. This module operates in conjunction with actuators or output devices, carrying out and overseeing the selected course of action. Feedback from the action module helps the planning module make necessary modifications and fixes.
- When a robotic arm is used in a factory, the action module gets instructions on how to pick up and move things. It then sends signals to the robot's motors to carry out the exact motions required to ensure precision assembly jobs.

5. Learning Strategies: The Engine of Adaptation

- The incorporation of learning strategies- mechanisms that let agents change, become better, and learn new things- is essential to agent architecture. Common methods include supervised learning, unsupervised learning, and reinforcement learning. Reinforcement learning directs agents' behavior toward the best results by providing feedback in the form of incentives or punishments. Labeled examples are necessary for supervised learning in order for the agent to anticipate or act. Conversely, unsupervised learning enables agents to find patterns and connections in data without specific direction.
- By improving the agent's capacity to draw conclusions from the past and adjust to new circumstances, these learning techniques help them improve their performance over time.

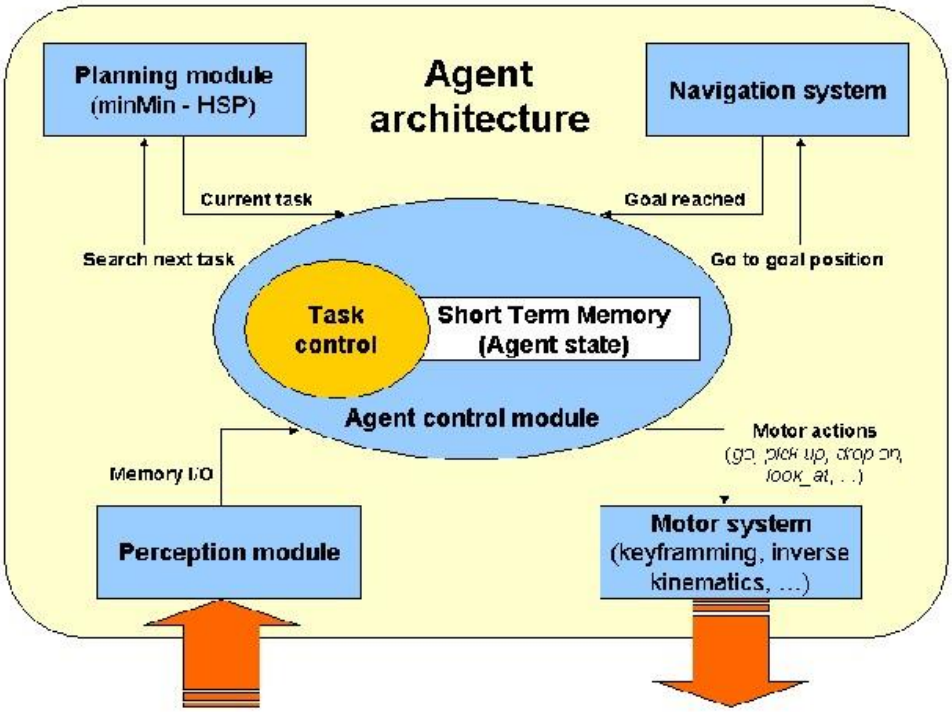


Fig: Agent Architecture

Conclusion

AI agent architectures offer a organized system for comprehending how smart systems independently perceive, reason, and make decisions in their surroundings. The components talked about profiling module, memory module, planning module, action module, and learning strategies- all work together to allow agents to engage with their environment, make choices, and adjust as needed. AI agents can reach higher levels of autonomy and intelligence by combining sensory skills, knowledge storage, decision-making abilities, action execution, and learning mechanisms. These structures act as crucial models for creating and assessing intricate AI systems, enabling researchers and engineers to create agents that can effectively complete particular tasks and objectives.

Types of Problem

There are basically three types of problem in artificial intelligence:

1. Ignorable: In which solution steps can be ignored.

2. Recoverable: In which solution steps can be undone.

3. Irrecoverable: Solution steps cannot be undo.

Steps problem-solving in AI: The problem of AI is directly associated with the nature of humans and their activities. So we need a number of finite steps to solve a problem which makes human easy works.

These are the following steps which require to solve a problem:

- **Problem definition:** Detailed specification of inputs and acceptable system solutions.
- **Problem analysis:** Analyze the problem thoroughly.
- **Knowledge Representation:** collect detailed information about the problem and define all possible techniques.
- **Problem-solving:** Selection of best techniques.

Components to formulate the associated problem:

- **Initial State:** This state requires an initial state for the problem which starts the AI agent towards a specified goal. In this state new methods also initialize problem domain solving by a specific class.
- **Action:** This stage of problem formulation works with function with a specific class taken from the initial state and all possible actions done in this stage.
- **Transition:** This stage of problem formulation integrates the actual action done by the previous action stage and collects the final stage to forward it to their next stage.
- **Goal test:** This stage determines that the specified goal achieved by the integrated transition model or not, whenever the goal achieves stop the action and forward into the next stage to determines the cost to achieve the goal.

- **Path costing:** This component of problem-solving numerical assigned what will be the cost to achieve the goal. It requires all hardware software and human working cost.

Problem Solving Agent

In artificial intelligence, a problem-solving agent refers to a type of intelligent agent designed to address and solve complex problems or tasks in its environment. These agents are a fundamental concept in AI and are used in various applications, from game-playing algorithms to robotics and decision-making systems. Here are some key characteristics and components of a problem-solving agent:

1. **Perception**: Problem-solving agents typically have the ability to perceive or sense their environment. They can gather information about the current state of the world, often through sensors, cameras, or other data sources.
2. **Knowledge Base**: These agents often possess some form of knowledge or representation of the problem domain. This knowledge can be encoded in various ways, such as rules, facts, or models, depending on the specific problem.
3. **Reasoning**: Problem-solving agents employ reasoning mechanisms to make decisions and select actions based on their perception and knowledge. This involves processing information, making inferences, and selecting the best course of action.
4. **Planning**: For many complex problems, problem-solving agents engage in planning. They consider different sequences of actions to achieve their goals and decide on the most suitable action plan.
5. **Actuation**: After determining the best course of action, problem-solving agents take actions to interact with their environment. This can involve physical actions in the case of robotics or making decisions in more abstract problem-solving domains.
6. **Feedback**: Problem-solving agents often receive feedback from their environment, which they use to adjust their actions and refine their problem-

solving strategies. This feedback loop helps them adapt to changing conditions and improve their performance.

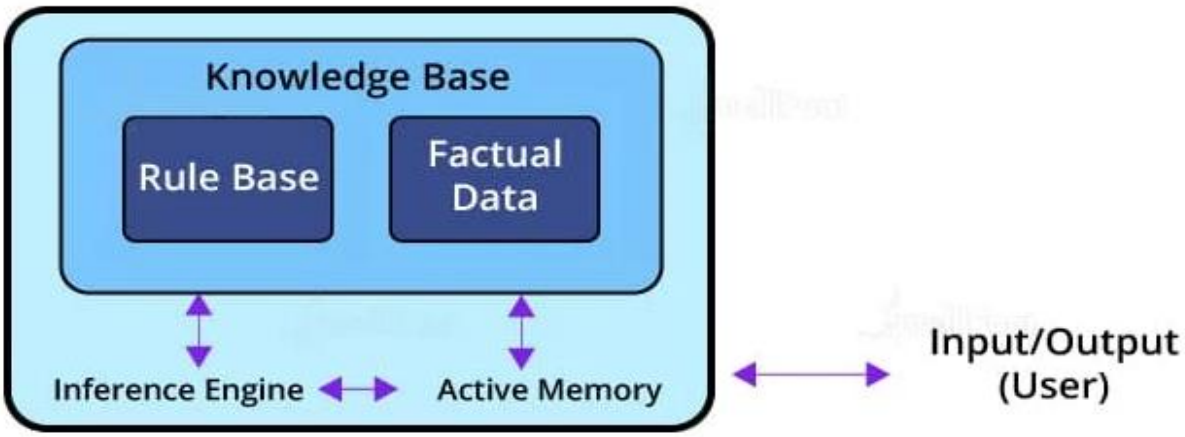
7. **Learning**: Some problem-solving agents incorporate machine learning techniques to improve their performance over time. They can learn from experience, adapt their strategies, and become more efficient at solving similar problems in the future.

Problem-solving agents can vary greatly in complexity, from simple algorithms that solve straight forward puzzles to highly sophisticated AI systems that tackle complex, real-world problems. The design and implementation of problem-solving agents depend on the specific problem domain and the goals of the AI application.

Production System in AI

A production system in AI is a framework that assists in developing computer programs to automate a wide range of tasks. It significantly impacts the creation of AI-based systems like computer software, mobile applications, and manufacturing tools. By establishing rules, a production system empowers machines to demonstrate particular behaviors and adapt to their surroundings.

In Artificial Intelligence, a production system serves as a cognitive architecture. It encompasses rules representing declarative knowledge, allowing machines to make decisions and act based on different conditions. Many expert systems and automation methodologies rely on the rules defined in production systems to guide their behavior.



A production system's architecture consists of rules structured as left-hand side (LHS) and right-hand side (RHS) equations. The LHS specifies the condition to be evaluated, while the RHS determines the output or action resulting from the estimated condition. This rule-based approach forms the foundation of production systems in AI, enabling machines to process information and respond accordingly.

Components of a Production System

For making an AI-based intelligent system that performs specific tasks, we need an architecture. The architecture of a production system in Artificial Intelligence consists of production rules, a database, and the control system.

Global Database

A global database consists of the architecture used as a central data structure. A database contains all the necessary data and information required for the successful completion of a task. It can be divided into two parts as permanent and temporary. The permanent part of the database consists of fixed actions, whereas the temporary part alters according to circumstances.

Production Rules

Production rules in AI are the set of rules that operate on the data fetched from the global database. Also, these production rules are bound with precondition and post condition that gets checked by the database. If a condition is passed through a production rule and gets satisfied by the global database, then the rule is successfully applied. The rules are of the form A®B, where the right-hand side represents an outcome corresponding to the problem state represented by the left-hand side.

Control System

The control system checks the applicability of a rule. It helps decide which rule should be applied and terminates the process when the system gives the correct output. It also resolves the conflict of multiple conditions arriving at the same time. The strategy of the control system specifies the sequence of rules that compares the condition from the global database to reach the correct result.

Characteristics of a Production System

There are mainly four characteristics of the production system in AI that is simplicity, modifiability, modularity, and knowledge-intensive.

Simplicity

The production rule in AI is in the form of an 'IF-THEN' statement. Every rule in the production system has a unique structure. It helps represent knowledge and reasoning in the simplest way possible to solve real-world problems. Also, it helps improve the readability and understanding of the production rules.

Modularity

The modularity of a production rule helps in its incremental improvement as the production rule can be in discrete parts. The production rule is made from a collection of information and facts that may not have dependencies unless there is a rule connecting them together. The addition or deletion of single information will not have a major effect on the output. Modularity helps enhance the performance of the production system by adjusting the parameters of the rules.

Modifiability

The feature of modifiability helps alter the rules as per requirements. Initially, the skeletal form of the production system is created. We then gather the requirements and make changes in the raw structure of the production system. This helps in the iterative improvement of the production system.

Knowledge-intensive

Production systems contain knowledge in the form of a human spoken language, i.e., English. It is not built using any programming languages. The knowledge is represented in plain English sentences. Production rules help make productive conclusions from these sentences.

Types of Production Systems

Production systems in AI can be categorized based on how they handle and process knowledge. This categorization includes Rule-Based Systems, Procedural Systems, and Declarative Systems, each possessing unique characteristics and applications.

1. Rule-Based Systems

i. Explanation of Rule-Based Reasoning

- Rule-based systems operate by applying a set of pre-defined rules to the given data to deduce new information or make decisions. These rules are generally in the form of conditional statements (if-then statements) that link conditions with actions or outcomes.

ii. Examples of Rule-Based Systems in AI

- **Diagnostic Systems**: Like medical diagnosis systems that infer diseases from symptoms.
- **Fraud Detection Systems**: Used in banking and insurance, these systems analyze transaction patterns to identify potentially fraudulent activities.

2. Procedural Systems

i. Description of Procedural Knowledge

- Procedural systems utilize knowledge that describes how to perform specific tasks. This knowledge is procedural in nature, meaning it focuses on the steps or procedures required to achieve certain goals or results.

ii. Applications of Procedural Systems

- **Manufacturing Control Systems**: Automate production processes by detailing step-by-step procedures to assemble parts or manage supply chains.
- **Interactive Voice Response (IVR) Systems**: Guide users through a series of steps to resolve issues or provide information, commonly used in customer service.

3. Declarative Systems

i. Understanding Declarative Knowledge

- Declarative systems are based on facts and information about what something is, rather than how to do something. These systems store knowledge that can be queried to make decisions or solve problems.

ii. Instances of Declarative Systems in AI

- **Knowledge Bases in AI Assistants**: Power virtual assistants like Siri or Alexa, which retrieve information based on user queries.

- **Configuration Systems**: Used in product customization, where the system decides on product specifications based on user preferences and declarative rules about product options.

Each type of production system offers different strengths and is suitable for various applications, from straightforward rule-based decision-making to complex systems requiring intricate procedural or declarative reasoning.

Game Playing

Game Playing is an important domain of artificial intelligence. Games don't require much knowledge; the only knowledge we need to provide is the rules, legal moves and the conditions of winning or losing the game. Both players try to win the game. So, both of them try to make the best move possible at each turn. Searching techniques like BFS (Breadth First Search) are not accurate for this as the branching factor is very high, so searching will take a lot of time. So, we need another search procedures that improve:

- **Generate procedure** so that only good moves are generated.
- **Test procedure** so that the best move can be explored first.

Game playing is a popular application of artificial intelligence that involves the development of computer programs to play games, such as chess, checkers, or Go. The goal of game playing in artificial intelligence is to develop algorithms that can learn how to play games and make decisions that will lead to winning outcomes.

1. One of the earliest examples of successful game playing AI is the chess program Deep Blue, developed by IBM, which defeated the world champion Garry Kasparov in 1997. Since then, AI has been applied to a wide range of games, including two-player games, multiplayer games, and video games.

There are two main approaches to game playing in AI, rule-based systems and machine learning-based systems.

1. **Rule-based systems** use a set of fixed rules to play the game.
2. **Machine learning-based systems** use algorithms to learn from experience and make decisions based on that experience.

In recent years, machine learning-based systems have become increasingly popular, as they are able to learn from experience and improve over time, making them well-suited for complex games such as Go. For example, AlphaGo, developed by DeepMind, was the first machine learning-based system to defeat a world champion in the game of Go.

Game playing in AI is an active area of research and has many practical applications, including game development, education, and military training. By simulating game playing scenarios, AI algorithms can be used to develop more effective decision-making systems for real-world applications. The most common search technique in game playing is **Minimax search procedure**. It is depth-first, depth-limited search procedure. It is used for games like chess and tic-tac-toe.

Minimax algorithm uses two functions –

MOVE GEN: It generates all the possible moves that can be generated from the current position.

STATIC EVALUATION: It returns a value depending upon the goodness from the viewpoint of two-player

This algorithm is a two player game, so we call the first player as PLAYER1 and second player as PLAYER2. The value of each node is backed-up from its children. For PLAYER1 the backed-up value is the maximum value of its children and for PLAYER2 the backed-up value is the minimum value of its children. It provides most promising move to PLAYER1, assuming that the PLAYER2 has make the best move. It is a recursive algorithm, as same procedure occurs at each level.

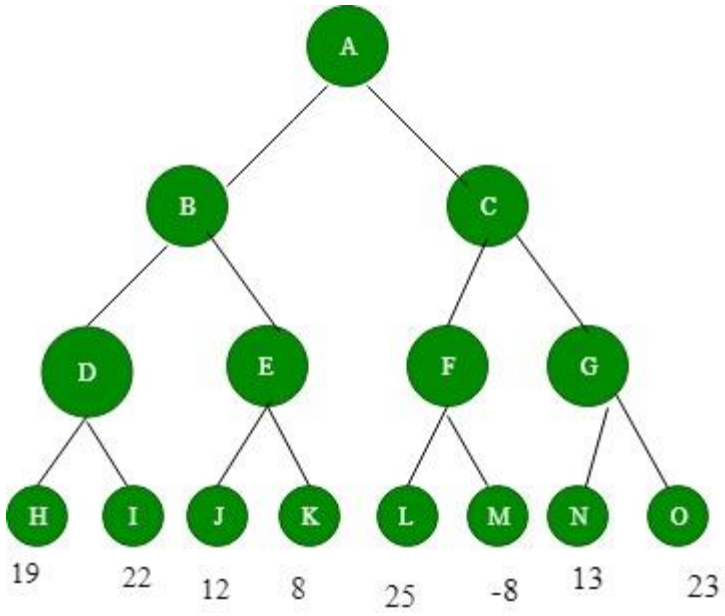


Figure 1: Before backing-up of values

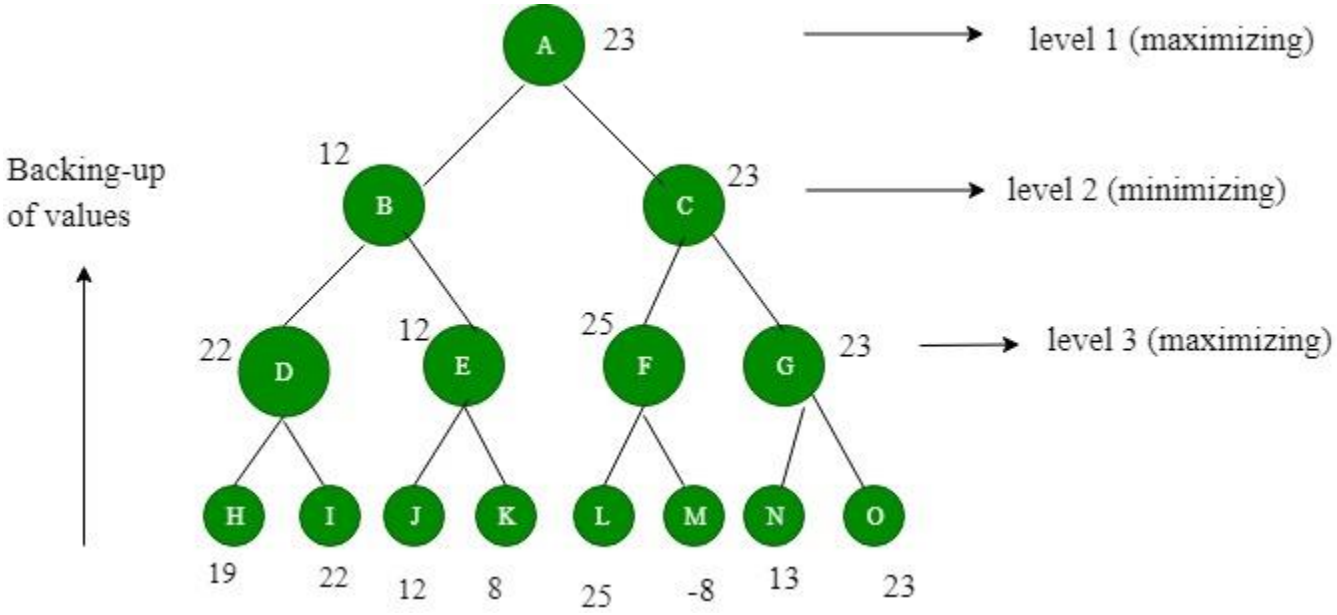


Figure 2: After backing-up of values

We assume that PLAYER1 will start the game, 4 levels are generated. The value to nodes H, I, J, K, L, M, N, O is provided by STATICEVALUATION function. Level 3 is maximizing level, so all nodes of level 3 will take maximum values of their children. Level 2 is minimizing level, so all its nodes will take minimum values of

their children. This process continues. The value of A is 23. That means A should choose C move to win.

Advantages of Game Playing in Artificial Intelligence:

1. **Advancement of AI:** Game playing has been a driving force behind the development of artificial intelligence and has led to the creation of new algorithms and techniques that can be applied to other areas of AI.
2. **Education and training:** Game playing can be used to teach AI techniques and algorithms to students and professionals, as well as to provide training for military and emergency response personnel.
3. **Research:** Game playing is an active area of research in AI and provides an opportunity to study and develop new techniques for decision-making and problem-solving.
4. **Real-world applications:** The techniques and algorithms developed for game playing can be applied to real-world applications, such as robotics, autonomous systems, and decision support systems.

Disadvantages of Game Playing in Artificial Intelligence:

1. **Limited scope:** The techniques and algorithms developed for game playing may not be well-suited for other types of applications and may need to be adapted or modified for different domains.
2. **Computational cost:** Game playing can be computationally expensive, especially for complex games such as chess or Go, and may require powerful computers to achieve real-time performance.

Problem-Solving Agents

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Problem-solving agents can vary greatly in complexity, from simple algorithms that solve straightforward puzzles to highly sophisticated AI systems that tackle complex, real-world problems. The design and implementation of problem-solving agents depend on the specific problem domain and the goals of the AI application.

Constraint Satisfaction Problems (CSP) in Artificial Intelligence

Finding a solution that meets a set of constraints is the goal of constraint satisfaction problems (CSPs), a type of AI issue. Finding values for a group of variables that fulfill a set of restrictions or rules is the aim of constraint satisfaction problems. For tasks including resource allocation, planning, scheduling, and decision-making, CSPs are frequently employed in AI.

There are mainly three basic components in the constraint satisfaction problem:

Variables: The things that need to be determined are variables. Variables in a CSP are the objects that must have values assigned to them in order to satisfy a particular set of constraints. Boolean, integer, and categorical variables are just a few examples of the various types of variables, for instance, could stand in for the many puzzle cells that need to be filled with numbers in a sudoku puzzle.

Domains: The range of potential values that a variable can have is represented by domains. Depending on the issue, a domain may be finite or limitless. For instance, in Sudoku, the set of numbers from 1 to 9 can serve as the domain of a variable representing a problem cell.

Constraints: The guidelines that control how variables relate to one another are known as constraints. Constraints in a CSP define the ranges of possible values for variables. Unary constraints, binary constraints, and higher-order constraints are only a few examples of the various sorts of constraints. For instance, in a sudoku problem, the restrictions might be that each row, column, and 3×3 box can only have one instance of each number from 1 to 9.

Constraint Satisfaction Problems (CSP) representation:

- The finite set of variables V1, V2, V3 ……..Vn.
- Non-empty domain for every single variable D1, D2, D3 ……..Dn.
- The finite set of constraints C1, C2 ……., Cm.
  - where each constraint Ci restricts the possible values for variables,
    - e.g., V1 ≠ V2
  - Each constraint Ci is a pair <scope, relation>
    - Example: <(V1, V2), V1 not equal to V2>
  - Scope = set of variables that participate in constraint.
  - Relation = list of valid variable value combinations.
    - There might be a clear list of permitted combinations. Perhaps a relation that is abstract and that allows for membership testing and listing.

Constraint Satisfaction Problems (CSP) algorithms:

- The **backtracking algorithm** is a depth-first search algorithm that methodically investigates the search space of potential solutions up until a solution is discovered that satisfies all the restrictions. The method begins by choosing a variable and giving it a value before repeatedly attempting to give values to the other variables. The method returns to the prior variable and tries a different value if at any time a variable cannot be given a value that fulfills the requirements. Once all assignments have been tried or a solution that satisfies all constraints has been discovered, the algorithm ends.
- The **forward-checking algorithm** is a variation of the backtracking algorithm that condenses the search space using a type of local consistency. For each unassigned variable, the method keeps a list of remaining values and applies local constraints to eliminate inconsistent values from these sets. The algorithm examines a variable's neighbors after it is given a value to see whether any of its remaining values become inconsistent and removes them from the sets if they do. The algorithm goes backward if, after forward checking, a variable has no more values.

- Algorithms for **propagating constraints** are a class that uses local consistency and inference to condense the search space. These algorithms operate by propagating restrictions between variables and removing inconsistent values from the variable domains using the information obtained.

Real-world Constraint Satisfaction Problems (CSP):

- **Scheduling:** A fundamental CSP problem is how to efficiently and effectively schedule resources like personnel, equipment, and facilities. The constraints in this domain specify the availability and capacity of each resource, whereas the variables indicate the time slots or resources.
- **Vehicle routing:** Another example of a CSP problem is the issue of minimizing travel time or distance by optimizing a fleet of vehicles' routes. In this domain, the constraints specify each vehicle's capacity, delivery locations, and time windows, while the variables indicate the routes taken by the vehicles.
- **Assignment:** Another typical CSP issue is how to optimally assign assignments or jobs to humans or machines. In this field, the variables stand in for the tasks, while the constraints specify the knowledge, capacity, and workload of each person or machine.
- **Sudoku:** The well-known puzzle game Sudoku can be modeled as a CSP problem, where the variables stand in for the grid's cells and the constraints specify the game's rules, such as prohibiting the repetition of the same number in a row, column, or area.
- **Constraint-based image segmentation:** The segmentation of an image into areas with various qualities (such as color, texture, or shape) can be treated as a CSP issue in computer vision, where the variables represent the regions and the constraints specify how similar or unlike neighboring regions are to one another.

Constraint Satisfaction Problems (CSP) benefits:

- conventional representation patterns
- generic successor and goal functions

- Standard heuristics (no domain-specific expertise).

Crypt-arithmetic Problems:

The **Crypt-Arithmetic problem in** Artificial Intelligence is a type of encryption problem in which the written message in an alphabetical form which is easily readable and understandable is converted into a numeric form which is neither easily readable nor understandable. In simpler words, the crypt-arithmetic problem deals with the converting of the message from the readable plain text to the non-readable cipher text. The constraints which this problem follows during the conversion is as follows:

1. A number 0-9 is assigned to a particular alphabet.
2. Each different alphabet has a unique number.
3. All the same, alphabets have the same numbers.
4. The numbers should satisfy all the operations that any normal number does.

Step to solving crypt-arithmetic problems:

1. **Identify Constraints**:
   - Each letter represents a unique digit (0-9).
   - Leading digits of numbers cannot be zero.
2. **Set Up the Problem**:
   - Write down the given equation clearly.
   - Note any obvious constraints, such as carrying over in addition.
3. **Trial and Error**:
   - Start with simpler letters that may have fewer possibilities.
   - Use logical deductions to limit the possibilities for each letter.
   - Apply trial and error to assign digits to letters, ensuring all constraints are satisfied.
4. **Check Your Solution**:
   - Ensure all letters are assigned a unique digit.
   - Verify the arithmetic equation holds true with your digit assignments.

Reference link: *Crypt-Arithmetic Problem - A type of Constraint Satisfactory Problem in Artificial Intelligence (includehelp.com)*

Example

Let us take an example of the message: SEND MORE MONEY.

Here, to convert it into numeric form, we first split each word separately and represent it as follows:

S E N D

M O R E

M O N E Y

These alphabets then are replaced by numbers such that all the constraints are satisfied. So initially we have all blank spaces.

These alphabets then are replaced by numbers such that all the constraints are satisfied. So initially we have all blank spaces. We first look for the MSB in the last word which is **'M'** in the word **'MONEY'** here. It is the letter which is generated by carrying. So, carry generated can be only one. SO, we have **M=1**.

Now, we have **S+M=O** in the second column from the left side. Here **M=1**. Therefore, we have, **S+1=O**. So, we need a number for **S** such that it generates a carry when added with **1**. And such a number is **9**. Therefore, we have **S=9** and **O=0**.

Now, in the next column from the same side we have E+O=N. Here we have O=0. Which means E+0=N which is not possible. This means a carry was generated by the lower place digits. So we have:

1+E=N ---------- (i)

Next alphabets that we have are N+R=E ------- (ii)

So, for satisfying both equations (i) and (ii), we get E=5 and N=6.

Now, R should be 9, but 9 is already assigned to S, So, R=8 and we have 1 as a carry which is generated from the lower place digits. Now, we have D+5=Y and this should generate a carry. Therefore, D should be greater than 4. As 5, 6, 8 and 9 are already assigned, we have D=7 and therefore Y=2.

Therefore, the solution to the given Crypt-Arithmetic problem is:

S=9; E=5; N=6; D=7; M=1; O=0; R=8; Y=2

Which can be shown in layout form as:

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  9 5 6 7
   1 0 8 5
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 1 0 6 5 2
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